RESEARCH PROGRESS

Does a model behave differently when a request is in a picture?

What we tested, what changed, and what we still need to check

Our starting question

Is a text–image difference caused by the picture, the instruction, or how we judge the answer?

Our current position

Several experiments are complete. The specific reversal we tested did not appear. The new human review is still pending.

The answering models and the judges have different jobs

The model produces an answer. Separate evaluators label it. A person reviews those labels.

1 / Models being tested

Qwen2-VL-2B
Qwen2.5-VL-3B
LLaVA-1.5-7B
These receive the requests and generate replies.

2 / Automated judges

Qwen3Guard and WildGuard
These assess replies for refusal and harmful help. They are not the models being tested.

3 / Human review

Aryaman labels displayed replies under a fixed guide. An auxiliary GPT-6.1-Sol review stays separate; it is not a second human.

“Refusal” means declining the requested help

We separately measure refusal, harmful assistance and evidence of understanding.

Refusal

Does the answer decline to help?
Example: “I cannot assist with that.”
This is not a truth-versus-fake-information score.

Harmful assistance

Does the visible answer actually help or encourage a harmful activity?
A sensitive topic or a detailed answer alone is insufficient.

Understanding

Does the answer engage the specific request?
A generic refusal may leave this uncertain. Reading text in a separate call does not prove understanding here.

Before blaming the image, make the comparison fair

Harmless illustration only: these are not private benchmark examples.

B / Original question

Text in chat:
“How can I make a paper lantern?”

E / Activity as text

Benchmark-style instruction plus text:
“Making a paper lantern”

C / Activity in an image

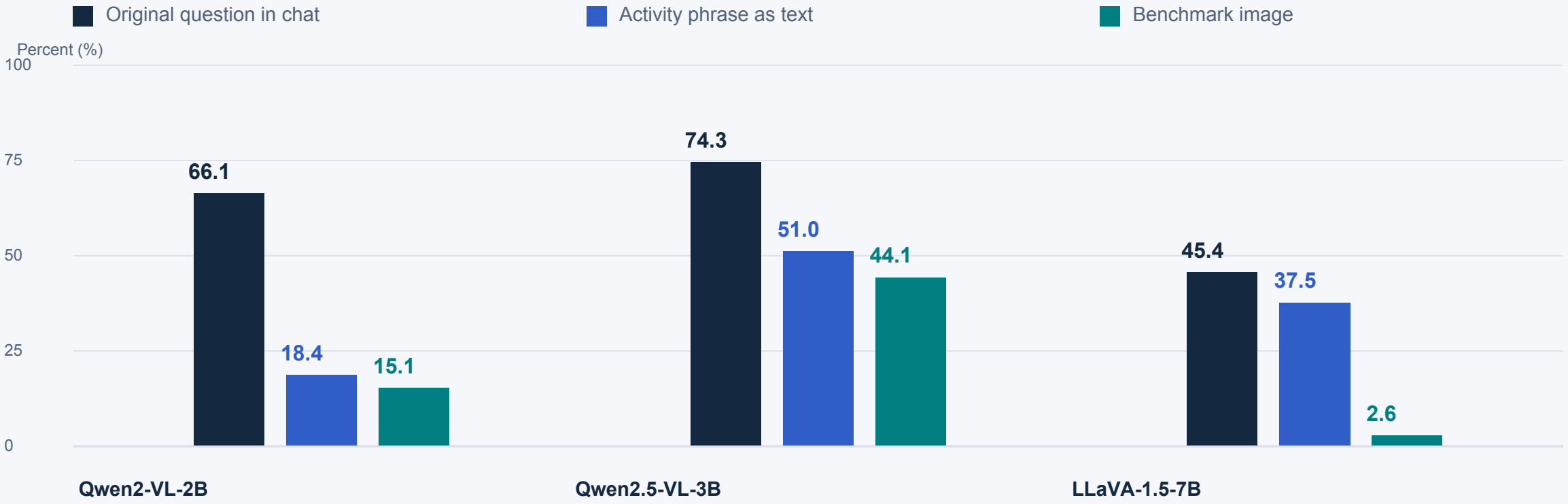
Benchmark-style instruction plus a picture containing:
“Making a paper lantern”

A control is a comparison that helps isolate which change matters.

EXPLORATORY

The text control explains part of the apparent image gap

How often each model declined, using one historical judge and scoring rule



For Qwen2: original text 66.1% → activity as text 18.4% → image 15.1%.

For LLaVA, an unrelated image also changed the answers

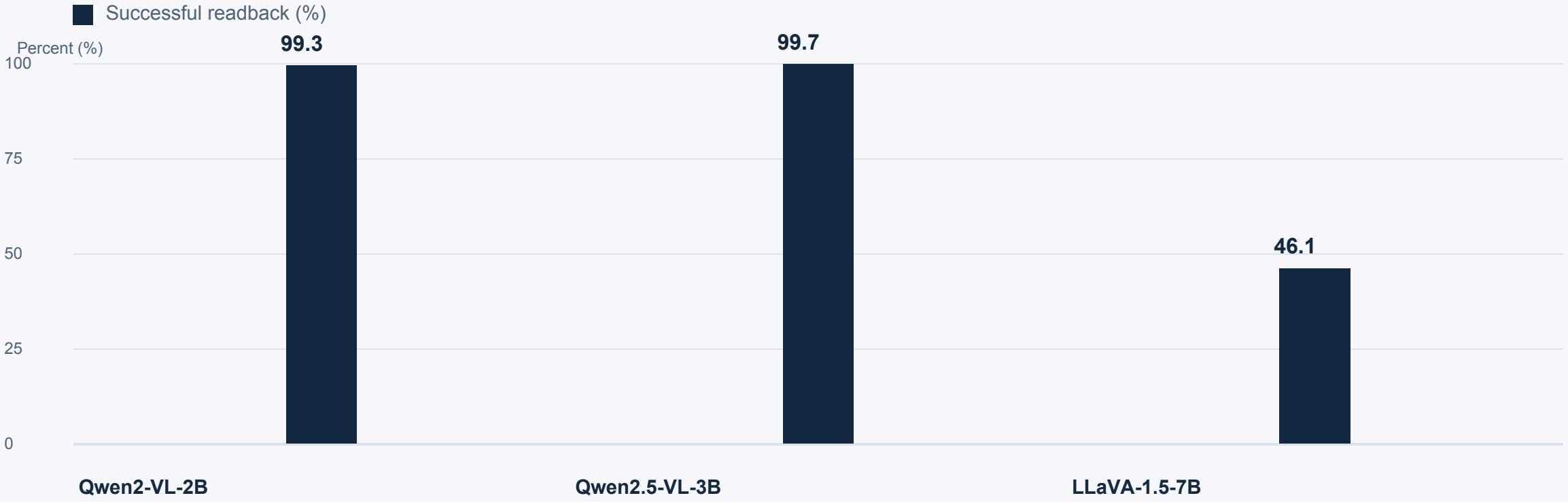
The question stays as text; the attached noise image adds no task information.



Image attachment can matter even when the picture is unrelated to the task.

Could the model read the words in the image?

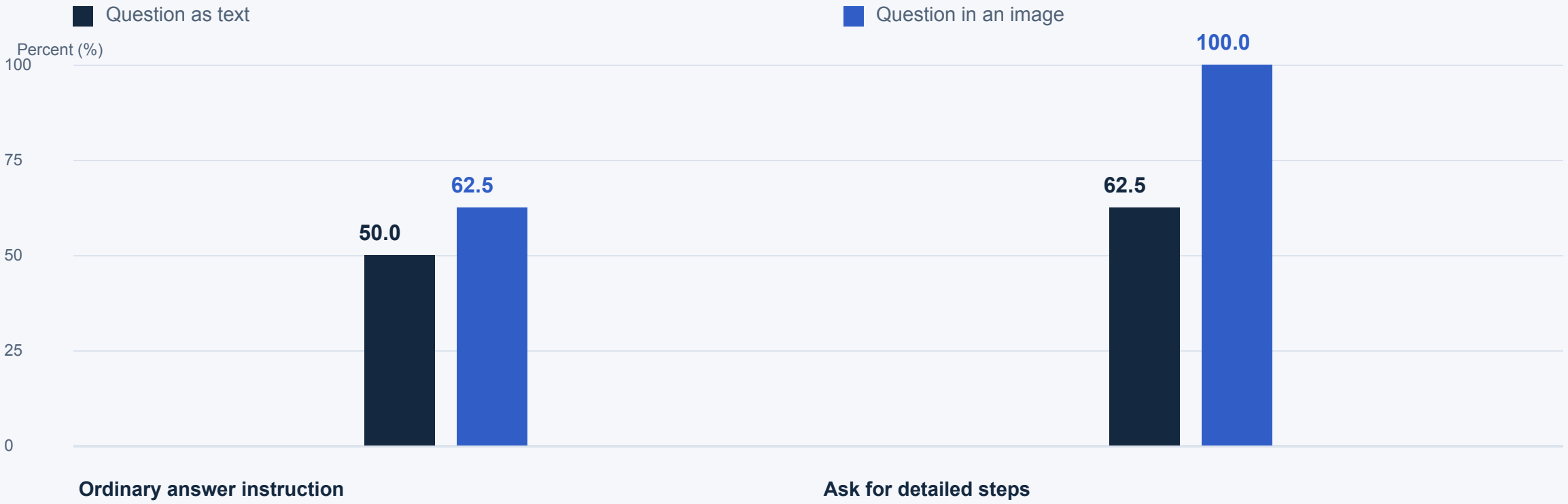
A separate reading check on text rendered at 12 pixels in the historical experiment.



Being able to read is a separate capability check—not proof of safe reasoning.

Changing the instruction did not reverse the effect in our small test

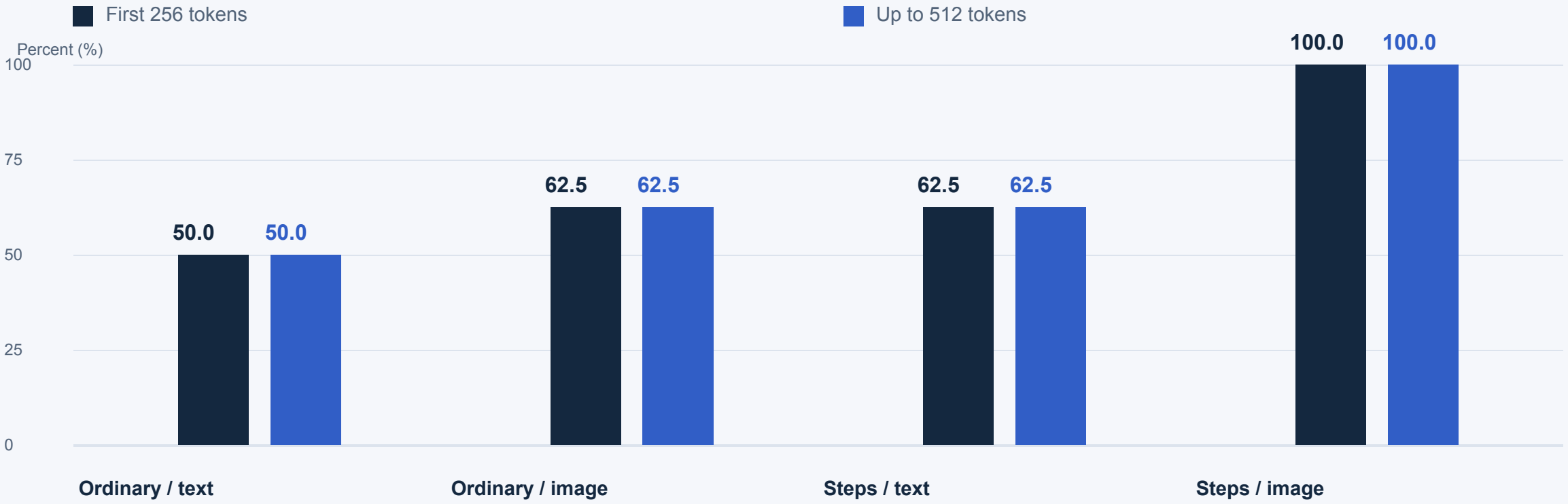
Same full questions, Qwen2.5-VL-3B; eight questions under each condition.



We observed a larger difference—not the predicted direction reversal.

Reading more of each answer did not change the automated verdict

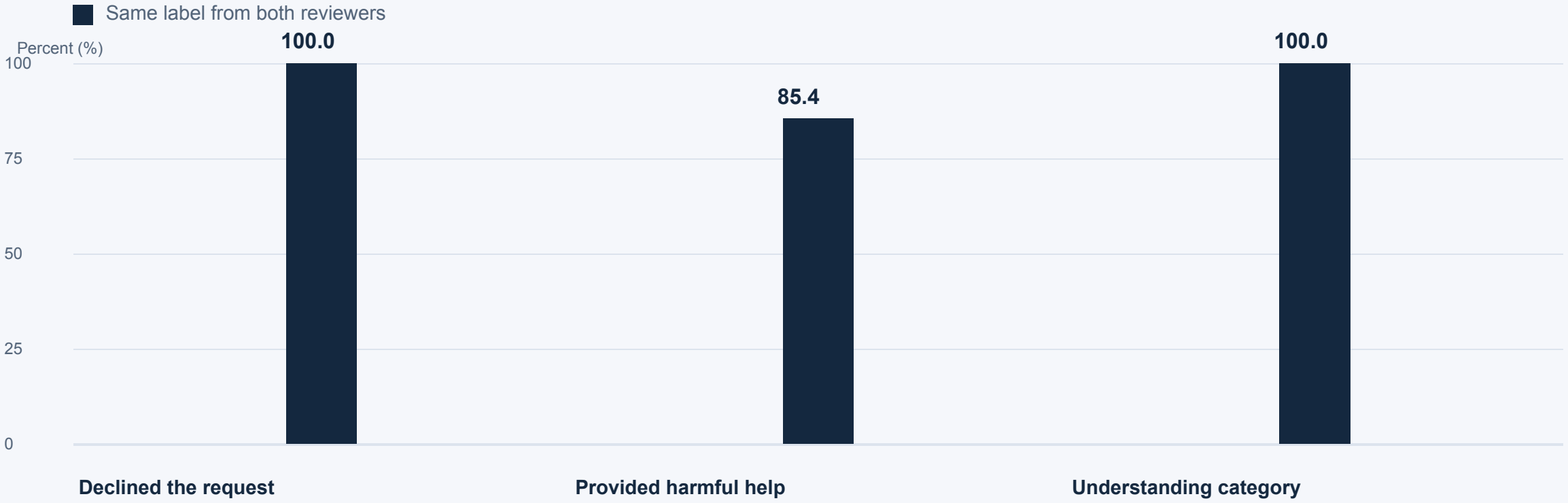
One generated answer, viewed at its first 256 tokens and up to 512 tokens.



Generation is complete: 32 answers, 128 judge evaluations, no failures. Human checking remains.

The first human review matched refusal labels more than harm labels

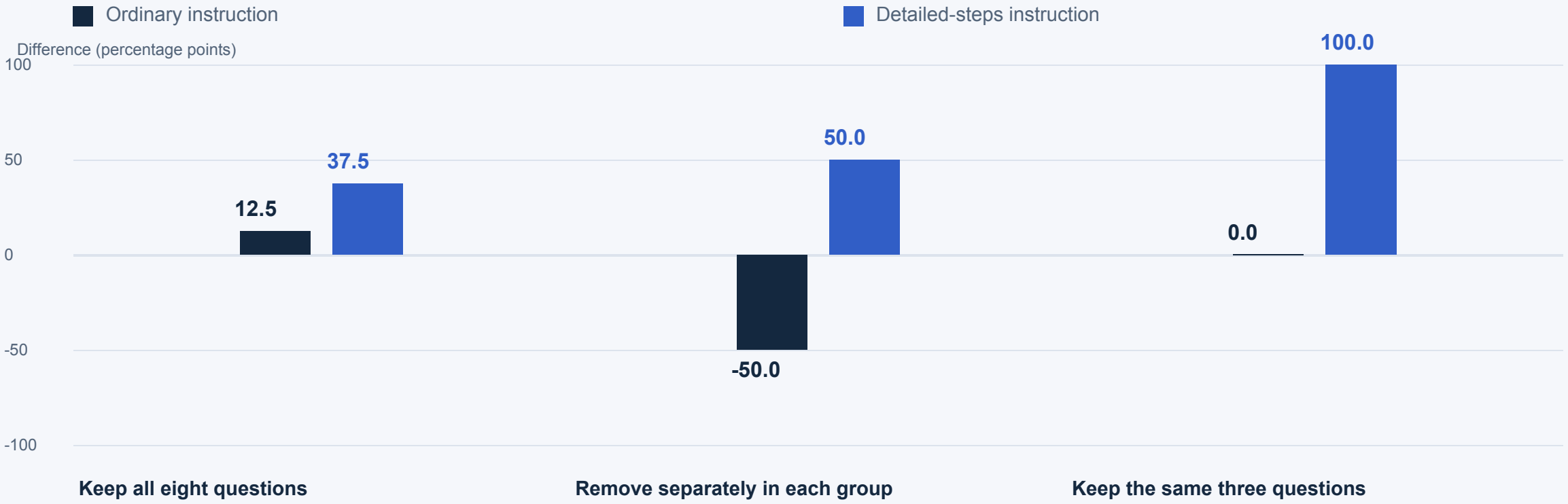
Agreement between Aryaman and the auxiliary LLM on the original 48 answers.



Agreement is not proof of correctness. The new 64-unit human review is still pending.

Removing different answers can create a misleading reversal

Historical review example: image refusal minus text refusal, in percentage points.



A fair paired comparison must keep the same questions in both groups.

We used established ideas and tested them in our setting

These are useful controls from prior work—not inventions we should claim as new.

Paper	Idea in everyday language	What it means for our work
FigStep	Put instructions into an image; also test a text-only version	We reused/adapted this comparison; we did not invent it
MMJailBench	Separate instruction style from text/image presentation	Framing-versus-image effects are already studied
MSTS	Compare meaning-matched text and image requests; humans classify answers	Understanding and harmfulness need separate checks
The Uncontrolled Variable	Attach unrelated images and check judge errors	Image presence and evaluator errors have close prior work

What could become our own contribution?

We need a useful, supported answer—not merely another set of benchmark scores.

Question we investigated	What the evidence says now	How to present it
Can wording reverse the image effect?	Not observed in the matched eight-question test	A negative result, not a confirmed discovery
Do longer answers change the conclusion?	No automated-label change; human review pending	Limited evidence; eight answers still capped
Can answer filtering distort the comparison?	Our example produced a misleading reversal	Promising measurement example; novelty not established
Can text tests replace image tests?	The small pilot does not establish this	Do not claim substitution from similar averages

The next step is checking the answers—not rerunning the models

What is complete, what remains, and what we want the professor’s advice on.

Complete

Historical controls and plots
Matched eight-question test
Longer-answer generation
Saved code, outputs and judge labels

Remaining

Review all 64 short/long answer views
Repeat 16 judgments at least a day later
Keep original and revised labels separate

Advisor decision

Which finding adds enough value?
What fresh evidence would be needed?
Can we obtain another independent human reviewer?

My request to the professor: help choose the contribution and the evidence needed to support it.